Arrival Theorem

$S(j)$ = Mean service time at device j

When there are $(n-1)$ customers, the mean number of customers in device j is z

Mean response time at device j when there are n customers = $(z + 1) S(j)$

Note "(n)" means there are n customers in th

$$\bar{n}_i(n) = \text{Mean \# of customers in}$$

$$R_i(n) = \text{Mean response time in d}$$

$$R_0(n) = \text{Mean response time of t}$$

$$X_i(n) = \text{Throughput of device i}$$

$$X_0(n) = \text{Throughput of the syster}$$

$\bar{n}_1(0) = 0 \qquad V_1 = 1 \qquad S_1 = 10$

$\bar{n}_2(0) = 0 \qquad V_2 = 0.5 \qquad S_2 = 15$

$\bar{n}_3(0) = 0 \qquad V_3 = 0.5 \qquad S_3 = 30$

$R_1(1) = (1 + \bar{n}_1(0)) S_1 = 10$

$R_2(1) = (1 + \bar{n}_2(0)) S_2 = 15$

$R_3(1) = (1 + \bar{n}_3(0)) S_3 = 30$

$R_0(1) = V_1 R_1(1) + V_2 R_2(1) + V_3 R_3(1)$

$= 1 * 10 + 0.5 * 15 + 0.5 * 30 = 32.5$

$X_0(1) = 1 / R_0(1) = 1 / 32.5$

$X_1(1) = V_1 X_0(1) = 10/32.5 = 0.0308$

$X_2(1) = V_2 X_0(1) = 0.5 / 32.5 = 0.0154$

$X_3(1) = V_3 X_0(1) = 0.5 / 32.5 = 1/65$

$\bar{n}_1(1) = R_1(1) X_1(1) = 10 * 0.0308 =$
0.308

$\bar{n}_2(1) = R_2(1) X_2(1) = 15 * 0.154 = 0.231$

$\bar{n}_3(1) = R_3(1) X_3(1) = 30 * 0.154 = 0.462$

$\bar{n}_1(1) = 0.3077$ $V_1 = 1$ $S_1 = 10$
$\bar{n}_2(1) = 0.2310$ $V_2 = 0.5$ $S_2 = 15$
$\bar{n}_3(1) = 0.4620$ $V_3 = 0.5$ $S_3 = 30$

$R_1(2) = (1 + \bar{n}_1(1)) S_1 = (1 + 0.3077) * 10 = 13.007$
$R_2(2) = (1 + \bar{n}_2(1)) S_2 = (1+0.2310)*15 = 18.46$
$R_3(2) = (1 + \bar{n}_3(1)) * S_3 = (1+0.462)*30 = 43.8$

$R_0(2) = V_1 R_1(2) + V_2 R_2(2) + V_3 R_3(2) = 1 * 13.007 + 0.5 * 18.46 + 0.5 * 43.8 = 44.23$

You can continue to find:
$X_0(2)$, $X_1(2)$, $X_2(2)$, $X_3(2)$
$\bar{n}_1(2)$, $\bar{n}_2(2)$, $\bar{n}_3(2)$

Scratch:

$\bar{n}_1(0) = 0 \qquad V_1 = 1 \qquad S_1 = 10$
$\bar{n}_2(0) = 0 \qquad V_2 = 0.5 \qquad S_2 = 15$
$\bar{n}_3(0) = 0 \qquad V_3 = 0.5 \qquad S_3 = 30$

$R_1(1) = (1 + \bar{n}_1(0)) S_1 = 10$
$R_2(1) = (1 + \bar{n}_2(0)) S_2 = 15$
$R_3(1) = (1 + \bar{n}_3(0)) S_3 = 30$

$R_0(1) = V_1 R_1(1) + V_2 R_2(1) + V_3 R_3(1)$
$= 1 * 10 + 0.5 * 15 + 0.5 * 30 = 32.5$
$X_0(1) = 1 / 32.5 = 0.0308$
$X_1(1) = V_1 X_0(1) = 1 * 0.0308 = 0.0308$
$X_2(1) = V_2 X_0(1) = 0.5 * 0.0308 = 0.0154$
$X_3(1) = V_3 X_0(1) = 0.5 * 0.0308 = 0.0154$
$\bar{n}_1(1) = R_1(1) X_1(1) = 10 * 0.0308 = 0.308$
$\bar{n}_2(1) = R_2(1) X_2(1) = 15 * 0.0154 = 0.231$
$\bar{n}_3(1) = R_3(1) X_3(1) = 30 * 0.0154 = 0.462$

$\bar{n}_1(1) = 0.3077$ $V_1 = 1$ $S_1 = 10$
$\bar{n}_2(1) = 0.2310$ $V_2 = 0.5$ $S_2 = 15$
$\bar{n}_3(1) = 0.4620$ $V_3 = 0.5$ $S_3 = 30$

$R_1(2) = (1 + \bar{n}_1(1))\, S_1 = (1 + 0.3077) * 10 = 13.077$
$R_2(2) = (1 + \bar{n}_2(1))\, S_2 = (1 + 0.2310) * 15 = 18.46$
$R_3(2) = (1 + \bar{n}_3(1))\, S_3 = (1 + 0.4620) * 30 = 43.8$

$R_0(2) = V_1\, R_1(2) + V_2\, R_2(2) + V_3\, R_3(2)$
$= 1 * 13.077 + 0.5 * 18.46 + 0.5 * 43.8 = 43.8$

You can continue to find:
$X_0(2)$, $X_1(2)$, $X_2(2)$, $X_3(2)$
$\bar{n}_1(2)$, $\bar{n}_2(2)$, $\bar{n}_3(2)$

Scratch:

$\bar{n}_1(0) =$
$\bar{n}_2(0) =$
$\bar{n}_3(0) =$

$R_1(1) =$
$R_2(1) =$
$R_3(1) =$

$R_0(1) =$

$X_0(1) =$

$X_1(1) =$
$X_2(1) =$
$X_3(1) =$

$\bar{n}_1(1) =$
$\bar{n}_2(1) =$
$\bar{n}_3(1) =$

$\bar{n}_1(1) = 0.3077$
$\bar{n}_2(1) = 0.2310$
$\bar{n}_3(1) = 0.4620$

$R_1(2) = (1 + \bar{n}_1(1))\ S_1 =$
$(1+0.0377)*10 = 13.08$
$R_2(2) = (1 + \bar{n}_2(1))\ S_2 = (1+0.2310)*15$
$= 18.46$
$R_3(2) = (1 + \bar{n}_3(1))\ S_3 = (1+0.4620)*30$
$= 43.8$

$R_0(2) = V_1\ R_1(2) + V_2\ R_2(2) + V_3\ R_3(2)$
$= 1 * 13.08 + 0.5 * 18.46 + 0.5 *$
43.8

$X_0(2) =$

$X_1(2) =$
$X_2(2) =$